



Revealing consumer hearts – the impact of emotional appeals in movie marketing: evidence from marketer-generated content on Weibo

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ABSTRACT

This study aims to investigate the effectiveness of emotional appeals in driving user engagement with marketer-generated content (MGC), specifically analyzing variations across the pre- and post-release phases of a movie. The study analyzed 23,998 marketing messages from Sina Weibo related to 356 movies released in China. The findings show that emotional appeals in MGC significantly enhance user engagement, with a stronger impact before a movie's release, especially when audiences lack prior knowledge about the movie. This study contributes to the marketing and film literature by empirically demonstrating the strategic importance of emotional appeals in MGC strategies, particularly during the pre-release period, in effectively capturing audience attention and fostering engagement.

1. Introduction

Various enterprises are generating content called marketer-generated content (MGC) about their products and services on social media to enhance user interaction and influence purchase intentions. Social-media content is typically categorized into user-generated (UGC) and MGC content based on the creator of the message. MGC encompasses marketing messages and promotional materials created by brand marketers, whereas UGC refers to content produced by users or consumers [1]. Specifically, MGC is considered a form of advertising that is entirely controlled by a firm and aligned with its marketing strategy [2]. Increasingly, businesses are leveraging MGC on social-media platforms—through text, images, and videos—to foster user engagement and influence purchase intentions. Incorporating effective MGC strategies enhances user engagement—a critical factor in outcomes such as product purchases and brand recognition [3–5]. In the experience-driven movie industry, where audiences have limited access to pre-release information beyond that provided by companies, MGC serves as a primary source shaping audience expectations and interest, underscoring its strategic importance. Despite the influence of MGC in shaping audience behavior, few studies have investigated MGC strategies aimed at optimizing user engagement in the movie industry.

Prior research shows that marketers employ diverse MGC strategies

to drive engagement and attract new consumers, including rational or emotional appeals [6–10], hedonic or utilitarian appeals [11], and directed or undirected communication [12]. Among these, emotional appeals have been extensively used across different brands and products [13,14]. Ji et al. [9] noted that the immediacy and personalization afforded by platforms such as Facebook, Instagram, and X (formally Twitter) created an environment conducive to the significant impact of emotional appeals on user engagement. Given the dynamic nature of social media, exploring how emotional appeals in MGC drive engagement metrics such as likes, comments, reposts, and overall user engagement is essential.

Nevertheless, previous studies show mixed findings on the role of emotional appeals within the context of brand and product marketing. While some studies have found a positive influence on user engagement [7–9,15,16], others have reported a negative one [6] or found no significant relationship [17–20]. These inconsistencies may arise from insufficient differentiation across product types and industry contexts, underscoring the need for an empirical examination of MGC in the movie industry. Accordingly, this study first examines whether emotional appeals in movie-related MGC are positively associated with user engagement.

Prior research suggests that emotional appeals are particularly persuasive when audiences have limited prior knowledge [21,22]. In the

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movie industry, where pre-release information is predominantly controlled by official channels, MGC emerges as a vital source. Building on this idea, and drawing on the heuristic-systematic model (HSM), we further investigate whether the effect of emotional appeals is stronger before a movie's release than after release. To achieve this, 23,998 movie-generated contents (MGCs) related to 356 movies released in China were analyzed using data from Sina Weibo. Regression analysis was employed to evaluate the effects of emotional appeals on user engagement over time, providing insights into the temporal dynamics of emotional marketing strategies.

This study has two major theoretical and practical contributions. First, it advances research on user engagement by empirically confirming that emotional appeals in movie-related MGC boost user engagement, providing practical guidelines for content creation. Second, it reveals that emotional appeals have a stronger impact before a movie's release. Guided by the HSM, this study shows that limited pre-release information leads users to process emotional content heuristically, suggesting that the movie industry should prioritize emotional MGC in the pre-release phase.

2. Literature review

2.1. Strategy of MGC: Emotional appeal

Given the interactive dynamics intrinsic to social-media platforms, marketers have implemented various MGC strategies. These encompass a spectrum of approaches, including rational or emotional appeals [6–10], hedonic or utilitarian appeals [11], and directed or undirected communication strategies [12]. The intent behind these strategies is not only to attract a larger user base but also to enhance overall user engagement on these platforms. Table 1 summarizes previous studies that classified MGC strategies and the impact of each type, covering a wide range of fields and platforms.

Among various MGC strategies, emotional appeals are more frequently used in social media marketing than other appeal strategies [13,14]. Emotional appeals rely on emotive words, images, or videos to elicit emotional responses, thereby capturing individual attention [24],

establishing profound emotional connections with consumers [25], and driving behaviors such as ad clicks [26], loyalty [27], and purchase intentions [28]. Comparative studies consistently show emotional appeals outperforming informational ones in boosting click-through rates, attitudes, and engagement [29,30,14,31].

Researchers have also explored various dimensions which type of emotions work best in social-media marketing. Positive appeals often increase favorable brand attitudes and purchase intentions [32], whereas negative emotions like sadness may be more persuasive in certain for-profit contexts [33]. Nonetheless, recent research has introduced nuanced insights, indicating that mixed emotional appeals hold greater persuasiveness compared to single positive appeals. This has been evidenced in their ability to enhance positive word-of-mouth (WOM) [34] and foster consumers' favorable evaluations of brands [35].

The effects of emotional appeals are further moderated by product type and user traits. Emotional messages are especially effective for experience goods and services [36–38,31], though some studies show the opposite in healthcare contexts [39]. Gender and involvement likewise moderate responses, with stronger effects for women and highly involved consumers [32,40,28]. Moreover, the characteristics of the posting account can influence outcomes. Recent research shows that emotional expressions by virtual influencers (VIs) on Instagram elicit even stronger engagement than those by human influencers (HIs), highlighting account type as a critical factor [41].

However, despite considerable research on emotional appeals, most research has focused on Western platforms, leaving cultural contexts like China underexplored. Similarly, how emotional appeals work across different phases of movie release has rarely been examined.

To fill these gaps, this study investigates how emotional appeals in movie MGC affect user engagement on Chinese social media, specifically on Weibo. Using sentiment analysis, this study extends the literature by offering insights into the relationship between emotional appeals and user engagement, particularly in the underexplored context of the Chinese movie market.

2.2. User engagement

Engagement denotes users' profound involvement in the media and their connection to the conveyed messages as an abstract concept [42]. The notion of user engagement originates from the concept of consumer-engagement behavior. Bowden [43] defined consumer engagement as customers' emotional response when they had a conscious or subconscious connection with products or brands, e.g., thinking about the brand or feeling good after watching the related advertisements.

User engagement functions as a pivotal catalyst in amplifying the reach and virality of MGC within the realm of social media marketing [44]. When users actively interact with content, particularly through actions such as sharing or reposting, they extend the visibility of the content to their personal networks. This ripple effect not only significantly broadens the audience base but also contributes to the development of brand loyalty, as evidenced by the positive correlation with higher company share values [45]. Moreover, researchers have proven that consumers' active online engagement with firm-hosted fan pages positively influences their perceptions of brands, consequently increasing the likelihood of offline store visits and influencing purchasing behaviors [46,47].

In addition, user engagement serves as a valuable metric for assessing the efficacy of a movie-marketing campaign. A comprehensive analysis of metrics such as likes, comments, shares, and overall interactions offers valuable insights into the resonance of MGC with the target audience. By understanding which content elicits the most engagement, movie marketers can refine their strategies, develop more targeted campaigns, and tailor future MGC to align with audience preferences. Significantly, heightened levels of user engagement have been proven to correlate with increased success in terms of box-office

Table 1
Summary of prior literature on mgc strategy with emotional appeals.

Studies	Data Source	Findings
Pinto and Yagnik [20]	Fitness tracker brands' official Facebook posts	Both appeal types had nonsignificant impacts on user engagement
Grover and Kar [18]	Mobile wallet firms' official tweets	
Dolan et al. [8]	12 wine brands' Facebook Insights and NCapture posts	Both appeal types positively impacted user engagement
Tan and Chen [15]	7 food companies' Wechat posts	
Araujo et al. [17]	Top 100 global brands' official tweets	Rational appeals positively impacted user engagement, while emotional appeals were nonsignificant
Lee and Hong [19]	Experiment using a car-sharing ad	
Ji et al. [9]	106 global brands' official Facebook posts	Emotional appeals had positive impacts on user engagement and purchase intention
Meire et al. [10]	A European soccer team's Facebook fan page	
Pletikosa Cvijikj and Michahelles [7]	100 food/beverage brands' official Facebook posts	
Tellis et al. [16]	79 global brands' official YouTube channel	
Tao et al. [23]	Experiment based on the environment of a peer-to-peer accommodation platform	
Ashley and Tuten [6]	28 global brands' official Facebook posts	Emotional appeals negatively impacted user engagement

performance [3–5], making it a key indicator for evaluating the overall impact of marketing efforts.

2.3. Heuristic-systematic model

Dual process theory proposes that individuals process information through two distinct operating modes [48]. System 1 operates automatically, impulsively, and intuitively, and it relies on rapid processing. In contrast, System 2 operates deliberately, thoughtfully, and reflectively, and it is based on slower processing. System 1 prompts quick and spontaneous actions through associative connections and motivation, while System 2 makes rapid behavioral decisions based on knowledge of values and the probability of possible outcomes [49,50]. These two systems require different cognitive abilities and levels of effort.

From the perspective of motivation and ability to process information, the HSM, proposed by Chen and Chaiken [51], is also a representative part of dual process theories. The HSM holds that people's information processes are jointly determined by heuristic and systematic modes [51]. Individuals who make quick decisions using minimal cognitive effort based on simple cues engage in heuristic processing; they prefer information that is easy to obtain and understand. In contrast, people considering sufficient and various factors and paying a higher cognitive cost before purchasing engage in systematic processing [51]. Specifically, people with high motivation and the ability to evaluate received messages are more likely to engage in systematic processes, whereas those with low motivation and ability are more likely to adopt the heuristic approach [52]. The HSM has been widely applied as a dual-process framework across diverse domains, from health and risk communication (e.g., [53,54]) to online review credibility (e.g., [55,56]) and digital advertising and endorsement effectiveness (e.g., [57]).

The HSM has also been applied in various studies to investigate the processing of emotional messages. Several studies have utilized the HSM to investigate how the receiver processes emotional messages. Kousta et al. [58] concluded that emotional words were processed faster than rational ones, implying that processing emotional messages requires less effort and capacity than rational information. Consequently, emotional messages are more likely to induce heuristic processing than systematic processing because the heuristic way requires less ability and cognitive effort. Moreover, insufficient prior knowledge has been revealed to determine the information-processing route and persuasiveness of emotional appeals. Averbeck et al. [21] evaluated the influence of prior information on the processing of health-related fear appeals based on the theoretical framework of the HSM. They showed that having prior knowledge about a message mitigated the effect of fear arousal and resulted in systematic processing.

In sum, the HSM provides a useful perspective on how prior knowledge and information availability shape responses to emotional messages. In this study, the HSM is employed specifically to motivate our second hypothesis regarding the pre- versus post-release difference, rather than as the overarching theoretical framework.

2.4. Sentiment analysis in social-media marketing

Sentiment analysis, a subfield of natural-language processing, has become instrumental in understanding individuals' opinions, attitudes, and emotions regarding specific products, events, or topics [59]. Various techniques—including lexicon-based approaches, machine-learning methods, and deep learning models—are employed to analyze sentiment across online platforms. Due to its effectiveness, sentiment analysis is widely applied to diverse domains, such as analyzing consumer reviews to gather comprehensive feedback for improving products and services [60], leveraging social-media MGC to optimize marketing strategies (e.g., [15]), monitoring online blogs and forums to detect and forecast trends [61], and evaluating news articles and related posts to identify fake news [62].

In the context of social media, sentiment analysis of UGC has been

utilized to evaluate consumers' attitudes toward brands [63], perceptions of service quality [64], the success of programs [65], election results [66], and business performance [67]. Additionally, from an individual perspective, sentiment analysis has been employed to detect public emotions, such as the threat and fear of terrorism [68], as well as individuals' depression levels [69].

When applied to MGC, sentiment analysis provides insights into the emotional tone and effectiveness of marketing messages. Prior studies have predominantly focused on exploring the relationship between the sentiment polarity of MGC and user engagement, revealing that emotionally charged posts are generally linked to higher levels of engagement, such as likes, shares, and favorable comments [70]. Interestingly, while positive sentiments may result in lower engagement levels [15], negative sentiments often garner increased attention and discussion through negative comment [71]. These insights enable marketers to craft emotionally engaging and targeted content, fostering brand loyalty and enhancing consumer interaction.

3. Hypotheses

User engagement is defined as users' behavioral participation, driven by motivation to engage actively [72]. On social media, engagement is commonly measured by time spent on a platform, interaction frequency and depth, and actions such as comments, shares, or clicks [8]. Following Moore et al. [73], this study defines emotional appeal as content that elicits an emotional response through emotive words. To measure emotional appeal, this study employs a lexicon-based sentiment analysis, using a predefined set of words or phrases (lexicons) associated with specific emotions (e.g., happiness, sadness, and anger) to count the number of emotional words within the text. For instance, in MGC, a message such as *“Stand for freedom, fight for change. This is not just a movie—a tribute to the fearless spirit that built our country. Be a part of the revolution on screen and let your heart swell with pride”* employs emotionally charged words to evoke patriotism, emphasizing themes of revolution and national pride.

Previous studies have demonstrated that emotional appeals in brand and product marketing enhance user engagement [7–9,15,16]. However, some studies have reported negative effects [6] or found no significant relationship [17–20]. These mixed findings may arise from varying product types and industry contexts. For example, previous studies have examined different industries, including fitness brands [20], mobile wallets [18], food and beverage [7], and car sharing [19].

To address this, the study focuses on an experience product—specifically movies—to examine whether emotional appeals in MGC positively impact social-media engagement. Research indicates that emotional appeals are especially effective for sensory experience products [36,74]. Drawing on Ji et al. [9], who found that stronger emotional content predicted more likes, comments, and shares on Facebook posts, this study hypothesizes that messages with higher emotional language correlate with increased user engagement:

Hypothesis 1. Emotional appeal in movie-related MGC is positively related to user engagement.

According to the HSM, prior knowledge shapes how individuals respond to emotional appeals because people with different levels of knowledge adopt distinct information-processing modes. When audiences lack prior knowledge, they are more likely to rely on heuristic cues, such as emotionally charged words, whereas those with greater prior knowledge are inclined to engage in systematic processing that draws on richer information sources [21,22].

In the movie industry, this distinction maps directly onto the release cycle. Before release, information about a movie is scarce and largely controlled by official marketing channels, which makes audiences more dependent on heuristic cues embedded in MGC. After release, however, abundant audience-generated content becomes available through online reviews and social media posts, enabling audiences to engage in more

systematic processing. This contextual shift implies that emotional appeals should exert stronger effects on user engagement before release, when audiences rely heavily on heuristics, than after release, when systematic processing predominates. Based on this reasoning, we expect that emotional appeals in pre-release marketing messages more strongly impact user engagement:

Hypothesis 2. *Emotional appeal in movie-related MGC has a stronger effect on user engagement before a movie's release than after.*

4. Methodology

4.1. Data collection

This study collected data on 549 movies released in China between January 2018 and December 2019. We focused on movies released before 2020 to avoid potential biases from disruptions caused by the COVID-19 pandemic, which significantly affected the movie industry from 2020 to 2022. Release date and each movie's screening period were collected from Maoyan (<https://www.piaofang.maoyan.com>), a Chinese movie-database site.

This study also collected marketing messages via Sina Weibo, which was founded in 2009 and has become China's most influential social-media platform. The number of moviegoers on Weibo hit 409 million in 2019, an increase of almost 50 % over 2018 [75], positioning Weibo as the most popular movie community in China. Therefore, we manually searched the official Weibo account for each movie and collected marketing messages from two weeks before to two weeks after each movie's release. Fig. 1 illustrates an example of MGC for movies on Weibo. After excluding 184 movies without official accounts and 9 re-releases, a total of 356 movies and 23,998 marketing messages were included in the analysis.

4.2. Variables

Table 2 presents the definitions of the variables. The dependent variable is user engagement, measured as the logarithm of a message's number of reposts (*Repost*), comments (*Comment*), and likes (*Like*). Gavilanes [76] conceptualized digital consumer engagement into four stages: viewing, liking, commenting, and sharing a post. Consistent with this framework, this study measures user engagement on Weibo by analyzing the frequency of likes, comments, and reposts.

The independent variable is the degree of emotional appeal (*Emotion*), measured as the logarithm of the number of emotional words in a message. This study encoded post-publish time as a dummy variable for the moderator of the second hypothesis, investigating the different impacts of emotional appeals before and after a movie's release; posts published before a movie's release are coded as 0, and those published after the release are coded as 1. Additionally, prior studies concluded that people tended to engage more with longer posts [77], and messages with hashtags or mentions were more likely to attract users [78]. Therefore, the number of words in a post (*Length*) and number of hashtags (#) and mentions (*HashMention*) were included as control variables. Finally, the variable, *Dummy_movie*, controls for all movie-level features.

To measure emotional appeals, this study counted the number of emotional words in each Weibo message. The first step in the sentiment-analysis process was to clean the texts by removing non-textual elements such as numbers, punctuation, and symbols. Thereafter, we used Python's Jieba package to segment each message into individual words, a common method for Chinese-text analysis (e.g., [79]). Following segmentation, the "cnsenti" algorithm was used to count emotional words in each message. This algorithm relies on "HowNet," a Chinese emotional-word lexicon developed by Dalian University of Technology. "HowNet" includes 27,466 Chinese emotional words and phrases, each annotated with information such as part of speech, emotion category,



从平凡少年到拯救世界的英雄,命运的编织者早已将线索埋下年度幻想冒险巨制#星河探梦:光辉之始#即将震撼上映!探索宇宙深处的秘密,追寻主角凯尔与神秘生物“星灵”之间的命运羁绊。一场意外的相遇,一段穿越星河的旅程,当遗失的文明重见天日,是选择守护,还是重塑未来?星河中的一切将如何改变?这个年末,一起踏上征程,见证光辉如何点亮宇宙!

Hide translation Translation from the Internet

From an ordinary boy to a hero destined to save the world, the threads of fate have already been woven. Epic Fantasy Adventure of the Year #Stellar Odyssey: Rise of the Radiance# is about to hit theaters! Embark on a journey to uncover the secrets of the cosmos and witness the fateful bond between the protagonist Kyle and the mysterious entity "Starlight." An accidental encounter, a journey across the galaxy—when a lost civilization reawakens, will the choice be to protect or reshape the future? How will everything among the stars change? This year-end, join the adventure and witness how radiance illuminates the universe!



Fig. 1. Example of MGC for Movies on Weibo.

Table 2
Definition of variables.

Variable	Operational definition	
Dependent variable	<i>Repost</i>	Take the logarithm of the number of reposts of a post
	<i>Comment</i>	Take the logarithm of the number of comments on a post
	<i>Like</i>	Take the logarithm of the number of likes of a post
Independent variable	<i>Emotion</i>	Take the logarithm of the number of emotional words in a post
Moderating variable	<i>Time</i>	0 = before the release, 1 = after the release
Control variable	<i>Length</i>	Number of words of a post
	<i>HashMention</i>	Number of hashtags and mentions of a post

intensity, and valence. It categorizes emotions into seven types: Fear, Anger, Joy, Sadness, Disgust, Surprise, and Good [80]. Using the "cnsenti" algorithm, the study identified words that were emotional, and

the total number of emotional words in each message was used as the measure of *Emotion*.

5. Results

5.1. Descriptive analysis

Table 3 presents the descriptive statistics for all the variables. The number of emotional words per message ranges from 0 to 108, with 84.42 % of the messages containing at least one emotional word. On average, marketers write 80 words and include 2 hashtags or mentions per post. All dependent and independent variables were log-transformed to correct for skewness.

5.2. Hypothesis testing

The regression results for H1 and H2 are displayed in Table 4. Models 1, 2, and 3 consider three types of user engagement, respectively: *Repost*, *Comment*, and *Like*. A post's *Length* positively impacted *Repost* ($B = 0.001, p < 0.001$) and *Comment* ($B = 0.0003, p < 0.001$); however, no significant impact was found on *Like* ($B = 0.000, p > 0.05$). *HashMention* positively affected the number of reposts, comments, and likes ($B_{Repost} = 0.049, p < 0.001$; $B_{Comment} = 0.032, p < 0.001$; $B_{Like} = 0.025, p < 0.001$). Regarding *Emotion*, the number of emotional words had a significant impact on user engagement ($B_{Repost} = 0.263, p < 0.001$; $B_{Comment} = 0.114, p < 0.001$; $B_{Like} = 0.146, p < 0.001$), thus supporting H1. Regarding the moderating variable, *Time*, posts published after a movie's release received more participation than those published before the movie's release ($B_{Repost} = 0.195, p < 0.001$; $B_{Comment} = 0.219, p < 0.001$; $B_{Like} = 0.270, p < 0.001$). Furthermore, the emotional words of posts made before a movie's release impacted user engagement more than those made after the movie's release ($B_{Repost} = -0.209, p < 0.001$; $B_{Comment} = -0.158, p < 0.001$; $B_{Like} = -0.151, p < 0.001$), supporting H2.

5.3. Robustness check

5.3.1. Alternative measurement of emotional appeals

The previous analysis measured the emotionality of a message by counting the number of emotional words, which might have limitations in fully understanding the message context. To assess the robustness of the main findings, this study also calculated a sentiment score (Sentiment) to provide a more nuanced analysis. While counting emotional words, modifiers such as degree adverbs, exclamation marks, and negative words were considered to adjust the sentiment score. Specifically, the sentiment score was calculated by the same Python package, "cnsenti," while the algorithm was based on another popular Chinese lexicon, "HowNet" [81], released in 1999; it contains 4,569 positive and 4,370 negative words and is widely used in Chinese-sentiment analysis (e.g., [82,83]).

The algorithm identifies positive and negative words in a sentence and sets each emotional word's default sentiment score to 1. Then, if there are any adverbs of degree near the emotional word, the sentiment score is multiplied by the corresponding weight, ranging from 0.5 to 4

Table 4
Regression results.

	Model 1 (DV = <i>Reposts</i>) Coef. (SE)	Model 2 (DV = <i>Comments</i>) Coef. (SE)	Model 3 (DV = <i>Likes</i>) Coef. (SE)
<i>Length</i>	0.001*** (0.0001)	0.0003*** (0.0001)	0.00005 (0.0001)
<i>HashMention</i>	0.049*** (0.002)	0.032*** (0.002)	0.025*** (0.002)
<i>Emotion</i>	0.263*** (0.02)	0.114*** (0.02)	0.146*** (0.02)
<i>Time</i>	0.195*** (0.02)	0.219*** (0.02)	0.270*** (0.01)
<i>Emotion × Time</i>	-0.209*** (0.03)	-0.158*** (0.03)	-0.151*** (0.02)
<i>Dummy_movie</i>	—	—	—
N	23,998	23,998	23,998
R ²	0.488	0.590	0.761
F	62.64***	94.28***	208.20***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table 5
Regression results of robustness check.

	Model 1 (DV = <i>Reposts</i>) Coef. (SE)	Model 2 (DV = <i>Comments</i>) Coef. (SE)	Model 3 (DV = <i>Likes</i>) Coef. (SE)
<i>Length</i>	0.0001*** (0.0001)	0.0003*** (0.0001)	0.0001* (0.0001)
<i>HashMention</i>	0.049*** (0.002)	0.032*** (0.002)	0.024*** (0.002)
<i>Sentiment</i>	0.072*** (0.01)	0.034*** (0.01)	0.036*** (0.01)
<i>Time</i>	0.186*** (0.02)	0.198*** (0.02)	0.262*** (0.01)
<i>Sentiment × Time</i>	-0.071*** (0.01)	-0.044*** (0.01)	-0.051*** (0.01)
<i>Dummy_movie</i>	—	—	—
N	23,998	23,998	23,998
R ²	0.487	0.590	0.760
F	62.17***	94.16***	207.73***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

points. Two points are added if there is an exclamation mark after the emotional word. If there are negative words (i.e., no, not, never, etc.), the score is multiplied by -1 ; however, because this study did not consider valance, the parameter was modified to 1. Consequently, emotional words were assigned different sentiment weights based on the context. Sentiment is the sum of the sentiment score for all emotional words. Sentiment ($M = 58.70$, $SD = 127.60$) is also right skewed; thus, it was log-transformed before regression analysis. The results were consistent with the initial findings, suggesting that emotional appeals positively influence user engagement, with a stronger effect for messages published before a movie's release (See Table 5).

5.3.2. Additional analysis for H2

The previous analysis concluded that emotional appeals had a time-varying effect on user engagement, with a greater impact on messages published before a movie's release than on those published afterward. An additional analysis for H2 was conducted by splitting the data into pre- and post-released posts to access the impact of emotional appeals. As shown in Table 6, for pre-released posts, emotional appeals positively influenced all types of user engagement. However, these effects diminished or disappeared for post-release posts. Specifically, the emotional-appeal coefficients for *Repost* and *Like* were smaller in post- than in pre-

Table 3
Descriptive statistics results of all variables.

Variable	Minimum	Maximum	Mean	S.D.	N
<i>Repost</i> ^a	0	5,668,471	1033.32	38914.49	23,998
<i>Comment</i> ^a	0	135,380	147.42	1489.76	23,998
<i>Like</i> ^a	0	834,469	1204.52	9545.87	23,998
<i>Emotion</i> ^a	0	108	2.84	3.11	23,998
<i>Time</i>	0	1	0.46	0.50	23,998
<i>Length</i>	0	2134	79.67	66.93	23,998
<i>HashMention</i>	0	50	2.19	2.25	23,998

^a Before taking the logarithm.

Table 6

Regression results of posts published before vs. after the movie's release.

	Model 1 (DV = <i>Repost</i>)			Model 2 (DV = <i>Comment</i>)			Model 3 (DV = <i>Like</i>)		
	Before Coef. (SE)	After Coef. (SE)	Z	Before Coef. (SE)	After Coef. (SE)	Z	Before Coef. (SE)	After Coef. (SE)	Z
<i>Length</i>	0.001*** (0.0001)	0.0003** (0.0001)		0.001*** (0.0001)	0.0003*** (0.0001)		0.0003** (0.0001)	0.000 (0.0001)	
<i>HashMention</i>	0.066*** (0.003)	0.028*** (0.003)		0.046*** (0.003)	0.016*** (0.003)		0.039*** (0.002)	0.007** (0.002)	
<i>Emotion</i>	0.215*** (0.03)	0.130*** (0.03)	2.00*	0.094*** (0.02)	0.010 (0.02)	32.88***	0.125*** (0.02)	0.049** (0.02)	2.69**
<i>Dummy_ movie</i>	—	—		—	—		—	—	
N	12,964	11,034		12,964	11,034		12,964	11,034	
R ²	0.479	0.570		0.570	0.671		0.740	0.824	
F	43.17***	39.82***		62.18***	61.26***		133.55***	140.55***	

*p < 0.05, **p < 0.01, *** p < 0.001.

release posts. Additionally, Z values ^[1] [84], which compare coefficient values between groups to examine effect-size differences, indicated that these differences were statistically significant, reinforcing the findings for H2. No significant relationship was found between *Emotion* and *Comment* ($B = 0.010$, $p > 0.05$).

5.3.3. Additional analysis sequels and franchise movies

Furthermore, as shown in Table 7, this study analyzed 48 sequels and franchise movies (e.g., Marvel movies), where certain information about a movie is available even before release. The results indicate that, for these movies as well, the impact of emotional appeals on user engagement is stronger before release than after.

6. Discussion

6.1. Findings

This study investigated effective strategies for movie-related MGC to foster user engagement, focusing on whether emotional appeal increased engagement and how this influence varied over time. We collected and analyzed 23,998 posts from Weibo accounts of 356 movies

Table 7

Regression Results (48 Series/franchise movies).

	Model 1 (DV = <i>Reposts</i>)	Model 2 (DV = <i>Comments</i>)	Model 3 (DV = <i>Likes</i>)
	Coef. (SE)	Coef. (SE)	Coef. (SE)
<i>Length</i>	0.002*** (0.0004)	0.002*** (0.0004)	0.0007* (0.0003)
<i>HashMention</i>	0.018* (0.008)	0.007 (0.007)	0.005 (0.005)
<i>Emotion</i>	0.261*** (0.07)	0.175** (0.06)	0.210*** (0.05)
<i>Time</i>	0.167** (0.06)	0.243*** (0.05)	0.274*** (0.04)
<i>Emotion × Time</i>	−0.386*** (0.10)	−0.329** (0.08)	−0.366*** (0.07)
<i>Dummy_movie</i>	—	—	—
N	2310	2310	2310
R ²	0.614	0.700	0.829
F	68.89***	101.41***	210.78***

*p < 0.05, **p < 0.01, *** p < 0.001.

¹ $Z = b1 - b2SE(b1)2 + SE(b2)2$, where b_n denotes the estimated coefficient, and $SE(b_n)$ denotes the estimated standard deviation of the group n .

released in China.

The results indicate that emotional appeals in movie-related MGC significantly boost user engagement, with a stronger effect observed before a movie's release. This finding is consistent with those in previous studies [7,9,16], which also found that emotional content in marketing messages enhanced user engagement. This variation in impact can be understood through the lens of the HSM, which suggests that limited prior information enhances the effect of emotional appeals.

6.2. Implications

This study has several academic and practical implications. First, it enhances our understanding of emotional appeals in movie MGC, specifically within the Chinese movie market. By analyzing a substantial dataset from Sina Weibo, this study provides empirical evidence that emotional appeals increase user engagement, highlighting the importance of emotional content in MGC strategies. This contributes to the academic literature by extending the relevance of emotional appeals beyond traditional advertising contexts to movie promotion.

Second, the study underscores that emotional appeals are particularly effective prior to a movie's release, emphasizing the importance of timing and strategic content integration. According to the HSM, individuals use heuristic processing with limited information, relying on emotional cues to make quick judgments. In movie marketing, the pre-release phase is marked by anticipation and curiosity, where heuristic processing dominates, and emotional appeals can effectively capture attention. This suggests that strategically integrating emotional content before release maximizes engagement and anticipation.

6.3. Limitations and future studies

The study has some limitations. First, its results may not be generalizable to other platforms or regions as it focuses specifically on Weibo and the Chinese movie market. Previous research supports this concern. For example, Bossetta and Schmökel [85] found that emotional-appeal effectiveness varied by platform. Their comparison of political candidates' images on Facebook and Instagram revealed that expressions of happiness performed better on Instagram, while calm expressions did better on Facebook, illustrating platform-specific differences. Similarly, Albers-Miller and Stafford [86] examined advertising appeals across Brazil, Taiwan, Mexico, and the United States (USA), finding that emotional and rational appeals differed based on both product type and cultural context, underscoring the role of culture in audience response. These studies suggest that differences in user behavior across platforms and cultural contexts may influence engagement. Accordingly, our findings should be interpreted as context-specific and exploratory rather than universally generalizable. To improve generalizability, future

research could replicate this study on other platforms, such as video-based platforms like YouTube and TikTok or globally oriented platforms such as X. This cross-platform and cross-cultural approach would clarify whether the findings extended beyond Weibo, thus enhancing their broader applicability.

Additionally, this study focused on text-based MGC rather than images or emojis to facilitate a more generalized sentiment analysis. While images offer a brief overview, text-based MGC allows for more nuanced and sustained engagement, which can foster enduring connections between audiences and movie content. Nevertheless, this text-based approach excludes symbols such as emojis, which prior research suggests are effective in conveying emotion [87,88]. To overcome these limitations, future research is necessary to explore the interaction between emojis and text, as well as images and text, in conveying emotions within movie-related MGC.

Last, this study did not differentiate among specific emotional categories (e.g., fear, joy, anger) when measuring emotional appeals, focusing instead on the overall degree of emotional wording. Future research could adopt more fine-grained approaches to disentangle the relative influence of different emotions and examine how their effectiveness may vary across contexts. In particular, prior studies suggest that movie trailers containing violent, sexual, or humorous content affect box-office performance [89]. Building on this evidence, future work could explore how the alignment between specific emotions in MGC and a movie's genre shapes user engagement, thereby offering more practical guidance for movie marketers.

7. Conclusion

This study investigated the role of emotional appeals in movie-related MGC and their effect on user engagement over time. Analyzing 23,998 Weibo posts, we found that emotional appeals significantly boosted engagement, particularly before a movie's release. This aligns with the HSM, underscoring the power of emotional content when prior information is limited.

The findings offer practical insights for movie marketers, suggesting that strategically timed emotional appeals before release can maximize engagement. Academically, this study broadens the understanding of emotional appeals in digital marketing and movie promotion.

However, the study has limitations, suggesting avenues for future research, such as replicating this approach on other platforms, including non-verbal indicators, and examining the interplay between emotions and movie genres. Overall, this study highlights the importance of emotional content and timing in effective MGC strategies.

CRedit authorship contribution statement

Zhien Piao: Conceptualization, Methodology, Data curation, Formal analysis, Writing – original draft. **Saerom Lee:** Conceptualization, Writing – review & editing, Validation. **Hyunmi Baek:** Supervision, Conceptualization, Methodology, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

References

- [1] K. Zhao, P. Zhang, H.M. Lee, Understanding the impacts of user-and marketer-generated content on free digital content consumption, *Decision Support Systems* 154 (2022) 113684, <https://doi.org/10.1016/j.dss.2021.113684>.
- [2] B. Schivinski, D. Dabrowski, The effect of social media communication on consumer perceptions of brands, *J. Mark. Commun.* 22 (2) (2016) 189–214, <https://doi.org/10.1080/13527266.2013.871323>.
- [3] A. Castillo, J. Benitez, J. Llorens, X.R. Luo, Social media-driven customer engagement and movie performance: theory and empirical evidence, *Decis. Support Syst.* 145 (2021) 113516, <https://doi.org/10.1016/j.dss.2021.113516>.
- [4] C. Ding, H.K. Cheng, Y. Duan, Y. Jin, The power of the 'like' button: the impact of social media on box office, *Decis. Support Syst.* 94 (2017) 77–84, <https://doi.org/10.1016/j.dss.2016.11.002>.
- [5] C. Oh, Y. Roumani, J.K. Nwankpa, H.F. Hu, Beyond likes and tweets: consumer engagement behavior and movie box office in social media, *Inf. Manag.* 54 (1) (2017) 25–37, <https://doi.org/10.1016/j.im.2016.03.004>.
- [6] C. Ashley, T. Tuten, Creative strategies in social media marketing: an exploratory study of branded social content and consumer engagement, *Psychol. Mark.* 32 (1) (2015) 15–27, <https://doi.org/10.1002/mar.20761>.
- [7] I.P. Cvijikj, F. Michahelles, Online engagement factors on Facebook brand pages, *Soc. Netw. Anal. Min.* 3 (4) (2013) 843–861, <https://doi.org/10.1007/s13278-013-0098-8>.
- [8] R. Dolan, J. Conduit, C. Frethey-Bentham, J. Fahy, S. Goodman, Social media engagement behavior: a framework for engaging customers through social media content, *Eur. J. Mark.* 53 (10) (2019) 2213–2243, <https://doi.org/10.1108/EJM-03-2017-0182>.
- [9] Y.G. Ji, Z.F. Chen, W. Tao, Z.C. Cathy Li, Functional and emotional traits of corporate social media message strategies: behavioral insights from S&P 500 Facebook data, *Public Relat. Rev.* 45 (1) (2019) 88–103, <https://doi.org/10.1016/j.pubrev.2018.12.001>.
- [10] M. Meire, K. Hewett, M. Ballings, V. Kumar, D. van den Poel, The role of marketer-generated content in customer engagement marketing, *J. Mark.* 83 (6) (2019) 21–42, <https://doi.org/10.1177/0022242919873903>.
- [11] J. Mu, J. Zhang, A. Borah, J. Qi, Creative appeals in firm-generated content and product performance, *Inf. Syst. Res.* 33 (1) (2022) 18–42, <https://doi.org/10.1287/isre.2021.1051>.
- [12] K.Y. Goh, C.S. Heng, Z. Lin, Social media brand community and consumer behavior: Quantifying the relative impact of user-and marketer-generated content, *Inf. Syst. Res.* 24 (1) (2013) 88–107, <https://doi.org/10.1287/isre.1120.0469>.
- [13] K. Swani, G.R. Milne, Evaluating Facebook brand content popularity for service versus goods offerings, *J. Bus. Res.* 79 (2017) 123–133, <https://doi.org/10.1016/j.jbusres.2017.06.003>.
- [14] K. Swani, G.R. Milne, B.P. Brown, A.G. Assaf, N. Donthu, What messages to post? evaluating the popularity of social media communications in business versus consumer markets, *Ind. Mark. Manag.* 62 (2017) 77–87, <https://doi.org/10.1016/j.indmarman.2016.07.006>.
- [15] S. Tan, W. Chen, How marketer-generated content characteristics affect consumer engagement? Empirical evidence from China's WeChat food marketing, *Br. Food J.* 124 (1) (2022) 255–274, <https://doi.org/10.1108/BFJ-12-2020-1169>.
- [16] G.J. Tellis, D.J. MacInnis, S. Tirunillai, Y. Zhang, What drives virality (sharing) of online digital content? the critical role of information, emotion, and brand prominence, *J. Mark.* 83 (4) (2019) 1–20, <https://doi.org/10.1177/0022242919841034>.
- [17] T. Araujo, P. Neijens, R. Vliegenthart, What motivates consumers to re-tweet brand content? the impact of information, emotion, and traceability on pass-along behavior, *J. Advert. Res.* 55 (3) (2015) 284–295, <https://doi.org/10.2501/JAR-2015-009>.
- [18] P. Grover, A.K. Kar, User engagement for mobile payment service providers—introducing the social media engagement model, *J. Retail. Consum. Serv.* 53 (2020) 101718, <https://doi.org/10.1016/j.jretconser.2018.12.002>.
- [19] J. Lee, I.B. Hong, Predicting positive user responses to social media advertising: the roles of emotional appeal, informativeness, and creativity, *Int. J. Inf. Manag.* 36 (3) (2016) 360–373, <https://doi.org/10.1016/j.ijinfomgt.2016.01.001>.
- [20] M.B. Pinto, A. Yagnik, Fit for life: a content analysis of fitness tracker brands use of Facebook in social media marketing, *J. Brand Manag.* 24 (1) (2017) 49–67, <https://doi.org/10.1057/s41262-016-0014-4>.
- [21] J.M. Averbeck, A. Jones, K. Robertson, Prior knowledge and health messages: an examination of affect as heuristics and information as systematic processing for fear appeals, *South Commun. J.* 76 (1) (2011) 35–54, <https://doi.org/10.1080/10417940902951824>.
- [22] P.C. Lewan, E. Stotland, The effects of prior information on susceptibility to an emotional appeal, *J. Abnorm. Soc. Psychol.* 62 (2) (1961) 450–453, <https://doi.org/10.1037/h0040869>.
- [23] D. Tao, W. Fang, B. Luo, L. Wan, Which Marketer-generated-content is more effective? an experimental study in the context of a peer-to-peer accommodation platform, *Int. J. Hosp. Manag.* 100 (2022) 103089, <https://doi.org/10.1016/j.ijhm.2021.103089>.
- [24] Google, 2015, Jun, "The First 5 Seconds: Creating YouTube Ads That Break Through in a Skippable World", Think with Google. <https://www.thinkwithgoogle.com/intl/en-apac/marketing-strategies/data-and-measurement/creating-youtube-ads-that-break-through-in-a-skippable-world/>.
- [25] D. Vrtana, A. Krizanova, "The power of emotional advertising appeals: Examining their influence on consumer purchasing behavior and brand-customer relationship, *Sustainability*" 15 (18) (2023) 13337, <https://doi.org/10.3390/su151813337>.

- [26] C. Wu, D. Sundiman, S.C. Kao, C.H. Chen, Emotion induction in click intention of picture advertisement: a field examination, *Journal of Internet Commerce* 17 (4) (2018) 356–382, <https://doi.org/10.1080/15332861.2018.1463803>.
- [27] S. Liengjindathaworn, K. Saenchaiyathon, P. Hawat, Effect of corporate reputation to brand loyalty of banks in Thailand, *Int. Bus. Manag.* 9 (1) (2015) 134–138, <https://doi.org/10.3923/ibm.2015.134.138>.
- [28] P. Jovanović, T. Vlastelica, S.C. Kostić, Impact of advertising appeals on purchase intention, *Journal of Sustainable Business and Management Solutions in Emerging Economies* 21 (81) (2017) 35–45, <https://doi.org/10.7595/management.fon.2016.0025>.
- [29] A. Japutra, F. Septianto, A.S. Can, Feeling grateful versus happy? the effects of emotional appeals in advertisements on self-made products, *J. Retail. Consum. Serv.* 69 (2022) 103091, <https://doi.org/10.1016/j.jretconser.2022.103091>.
- [30] J. Martín-Quevedo, E. Fernández-Gómez, F. Segado-Boj, How to engage with younger users on Instagram: a comparative analysis of HBO and Netflix in the Spanish and US markets, *Int. J. Media Manag.* 21 (2) (2019) 67–87, <https://doi.org/10.1080/14241277.2019.1585355>.
- [31] T. Teichert, D. Hardeck, Y. Liu, R. Trivedi, How to implement informational and emotional appeals in print advertisements: a framework for choosing ad appeals based on advertisers' objectives and targeted demographics, *J. Advert. Res.* 58 (3) (2018) 363–379, <https://doi.org/10.2501/JAR-2017-054>.
- [32] N. Dens, P. De Pelsmacker, Consumer response to different advertising appeals for new products: the moderating influence of branding strategy and product category involvement, *J. Brand Manag.* 18 (2010) 50–65, <https://doi-org-ssl.oca.korea.ac.kr/10.1057/bm.2010.22>.
- [33] I. Roozen, The impact of emotional appeal and the media context on the effectiveness of commercials for not-for-profit and for-profit brands, *J. Mark. Commun.* 19 (3) (2013) 198–214, <https://doi-org-ssl.oca.korea.ac.kr/10.1080/13527266.2011.619555>.
- [34] S. Quach, F. Septianto, P. Thaichon, T.M. Chiew, Mixed emotional appeal enhances positive word-of-mouth: the moderating role of narrative person, *J. Retail. Consum. Serv.* 62 (2021) 102618, <https://doi-org-ssl.oca.korea.ac.kr/10.1016/j.jretconser.2021.102618>.
- [35] B. Xin, C. Zhu, F. Septianto, The effects of mixed emotional appeals in leveraging paradox brands, *Journal of Business Research* 153 (2022) 266–275, <https://doi-org-ssl.oca.korea.ac.kr/10.1016/j.jbusres.2022.08.020>.
- [36] Y.C. Lin, Y.C. Lee, N.T. Lin, The effect of advertising using advertising appeals, pictures and product categories, *Journal of Statistics and Management Systems* 17 (1) (2014) 71–96, <https://doi.org/10.1080/09720510.2013.867716>.
- [37] H. Zhang, J. Sun, F. Liu, G. Knight, J.g., Be rational or be emotional: advertising appeals, service types and consumer responses, *European Journal of Marketing* 48 (11/12) (2014) 2105–2126, <https://doi.org/10.1108/EJM-10-2012-0613>.
- [38] L. Sudbury-Riley, L. Edgar, Why older adults show preference for rational over emotional advertising appeals: a UK brand study challenges the applicability of socioemotional selectivity theory to advertising, *J. Advert. Res.* 56 (4) (2016) 441–455, <https://doi-org-ssl.oca.korea.ac.kr/10.2501/JAR-2016-048>.
- [39] M.J. van Der Goot, N. Bol, J.C.V. Weert, Age differences in preferences for emotionally-meaningful versus knowledge-related appeals, *Communications* 46 (2) (2021) 205–228, <https://doi.org/10.1515/commun-2019-0108>.
- [40] Y.A. Jeon, Y. Ryoo, H.J. Yoon, Increasing the efficacy of emotional appeal ads on online video-watching platforms: the effects of goals and emotional approach tendency on ad-skipping behavior, *J. Advert.* (2022) 1–18, <https://doi.org/10.1080/00913367.2022.2073299>.
- [41] C. Ko, H. Baek, Effects of emotional expression on user engagement in virtual influencers' Instagram posts: a comparative analysis with human influencers, *J. Retail. Consum. Serv.* 88 (2026) 104484, <https://doi.org/10.1016/j.jretconser.2025.104484>.
- [42] B.J. Calder, E.C. Malthouse, Media engagement and advertising effectiveness, *Kellogg on Advertising and Media* 1 (2008) 1–36, <https://doi.org/10.1002/9781119198154.ch1>.
- [43] J.L. Bowden, The process of customer engagement: a conceptual framework, *J. Mark. Theory Pract.* 17 (1) (2009) 63–74, <https://doi.org/10.2753/MTP1069-6679170105>.
- [44] S. Chandrasekaran, B. Annamalai, S.K. De, Evaluating marketer generated content popularity on brand fan pages – a multilevel modelling approach, *Telematics Inform.* 44 (2019) e101266, <https://doi.org/10.1016/j.tele.2019.101266>.
- [45] İ.E. Erdoğan, M. Çiçek, “The impact of social media marketing on brand loyalty”, *Procedia – Soc. Behav. Sci.* 58 (2012) 1353–1360, <https://doi.org/10.1016/j.sbspro.2012.09.1119>.
- [46] M.R. Habibi, M. Laroche, M.O. Richard, The roles of brand community and community engagement in building brand trust on social media, *Comput. Hum. Behav.* 37 (2014) 152–161, <https://doi.org/10.1016/j.chb.2014.04.016>.
- [47] R. Rishika, A. Kumar, R. Janakiraman, R. Bezawada, The effect of customers' social media participation on customer visit frequency and profitability: an empirical investigation, *Inf. Syst. Res.* 24 (1) (2013) 108–127, <https://doi.org/10.1287/isre.1120.0460>.
- [48] S. Chaiken, Heuristic versus systematic information processing and the use of source versus message cues in persuasion, *J. Pers. Soc. Psychol.* 39 (5) (1980) 752, <https://doi.org/10.1037/0022-3514.39.5.752>.
- [49] E.R. Smith, J. DeCoster, Dual-process models in social and cognitive psychology: Conceptual integration and links to underlying memory systems, *Pers. Soc. Psychol. Rev.* 4 (2) (2000) 108–131, <https://doi.org/10.1207/S15327957PSPR0402.01>.
- [50] F. Strack, R. Deutsch, Reflective and impulsive determinants of social behavior, *Pers. Soc. Psychol. Rev.* 8 (3) (2004) 220–247, <https://doi.org/10.1207/s15327957pspr0803.1>.
- [51] S. Chen, S. Chaiken, The heuristic-systematic model in its broader context, in: S. Chaiken, Y. Trope (Eds.), *Dual-Process Theories in Social Psychology*, Guilford Press, New York, 1999, pp. 73–96.
- [52] K. Hallahan, Enhancing motivation, ability, and opportunity to process public relations messages, *Public Relat. Rev.* 26 (4) (2000) 463–480, [https://doi.org/10.1016/S0363-8111\(00\)00059-X](https://doi.org/10.1016/S0363-8111(00)00059-X).
- [53] S.K. Steginga, S. Occhipinti, The application of the heuristic-systematic processing model to treatment decision making about prostate cancer, *Med. Decis. Making* 24 (6) (2004) 573–583, <https://doi.org/10.1177/0272989X042710>.
- [54] C.W. Trumbo, Heuristic-systematic information processing and risk judgment, *Risk Anal.* 19 (3) (1999) 391–400, <https://doi.org/10.1111/j.1539-6924.1999.tb00415>.
- [55] H. Baek, J. Ahn, Y. Choi, Helpfulness of online consumer reviews: readers' objectives and review cues, *Int. J. Electron. Commer.* 17 (2) (2012) 99–126, <https://doi.org/10.2753/JEC1086-4415170204>.
- [56] K.Z. Zhang, S.J. Zhao, C.M. Cheung, M.K. Lee, Examining the influence of online reviews on consumers' decision-making: a heuristic-systematic model, *Decision Support Systems* 67 (2014) 78–89, <https://doi.org/10.1016/j.dss.2014.08.005>.
- [57] Y. Liu, K. Lei, The impact of human and virtual endorsements on advertising effectiveness: a comparative analysis based on different information appeals, *Humanities and Social Sciences Communications* 12 (2025) 177, <https://doi.org/10.1057/s41599-025-04509-4>.
- [58] S.T. Kousta, D.P. Vinson, G. Vigliocco, Emotion words, regardless of polarity, have a processing advantage over neutral words, *Cognition* 112 (3) (2009) 473–481, <https://doi.org/10.1016/j.cognition.2009.06.007>.
- [59] W. Medhat, A. Hassan, H. Korashy, Sentiment analysis algorithms and applications: a survey, *Ain Shams Eng. J.* 5 (4) (2014) 1093–1113, <https://doi.org/10.1016/j.asej.2014.04.011>.
- [60] R. Ireland, A. Liu, Application of data analytics for product design: sentiment analysis of online product reviews, *CIRP J. Manuf. Sci. Technol.* 23 (2018) 128–144, <https://doi.org/10.1016/j.cirpj.2018.06.003>.
- [61] N. Li, D.D. Wu, Using text mining and sentiment analysis for online forums hotspot detection and forecast, *Decis. Support Syst.* 48 (2) (2010) 354–368, <https://doi.org/10.1016/j.dss.2009.09.003>.
- [62] M.A. Alonso, D. Vilares, C. Gómez-Rodríguez, J. Vilares, Sentiment analysis for fake news detection, *Electronics* 10 (11) (2021) 1348, <https://doi.org/10.3390/electronics10111348>.
- [63] V. Ikoru, M. Sharmina, K. Malik, R. Batista-Navarro, Analyzing sentiments expressed on Twitter by UK energy company consumers, in: *In 2018 Fifth International Conference on Social Networks Analysis, Management and Security*, 2018, pp. 95–98.
- [64] L. Martín-Domingo, J.C. Martín, G. Mandsberg, Social media as a resource for sentiment analysis of Airport Service Quality (ASQ), *J. Air Transp. Manag.* 78 (2019) 106–115, <https://doi.org/10.1016/j.jairtraman.2019.01.004>.
- [65] S. Yuliyanti, T. Djatna, H. Sukoco, Sentiment mining of community development program evaluation based on social media, *Telecommunication Computing Electronics and Control* 15 (4) (2017) 1858–1864, <http://doi.org/10.12928/telekonnika.v15i4.4633>.
- [66] H. Wang, D. Can, A. Kazemzadeh, F. Bar, S. Narayanan, A system for real-time twitter sentiment analysis of 2012 us presidential election cycle, in: *In Proceedings of the ACL 2012 System Demonstrations*, 2012, pp. 115–120.
- [67] M.H. Saragih, A.S. Girsang, Sentiment analysis of customer engagement on social media in transport online, in: *In 2017 International Conference on Sustainable Information Engineering and Technology*, 2017, pp. 24–29.
- [68] S. Mansour, Social media analysis of user's responses to terrorism using sentiment analysis and text mining, *Procedia Comput. Sci.* 140 (2018) 95–103, <https://doi.org/10.1016/j.procs.2018.10.297>.
- [69] A.U. Hassan, J. Hussain, M. Hussain, M. Sadiq, S. Lee, Sentiment analysis of social networking sites (SNS) data using machine learning approach for the measurement of depression, in: *In 2017 International Conference on Information and Communication Technology Convergence*, 2017, pp. 138–140.
- [70] L. Ma, W. Ou, C.S. Lee, Investigating consumers' cognitive, emotional, and behavioral engagement in social media brand pages: a natural language processing approach, *Electron. Commer. Res. Appl.* 54 (2022) 101179, <https://doi.org/10.1016/j.eleap.2022.101179>.
- [71] K.K. Aldous, J. An, B.J. Jansen, What really matters? Characterising and predicting user engagement of news postings using multiple platforms, sentiments and topics, *Behav. Inform. Technol.* 42 (5) (2023) 545–568, <https://doi.org/10.1080/0144929X.2022.2030798>.
- [72] P.M. Di Gangi, M.M. Wasko, Social media engagement theory: Exploring the influence of user engagement on social media usage, *Journal of Organizational and End User Computing* 28 (2) (2016) 53–73, <https://doi.org/10.4018/JOEUC.2016040104>.
- [73] D.J. Moore, W.D. Harris, H.C. Chen, Affect intensity: an individual difference response to advertising appeals, *J. Consum. Res.* 22 (2) (1995) 154–164, <https://doi.org/10.1086/209442>.
- [74] K. Swani, G.R. Milne, P. Brown, B.p., Spreading the word through likes on Facebook, *J. Res. Interact. Mark.* 7 (4) (2013) 269–294, <https://doi.org/10.1108/JRIM-05-2013-0026>.
- [75] Weibo, 2019, “Weibo movie marketing white paper”, available at: <https://data.weibo.com/report/reportDetail?id=443> (accessed 24 October 2020).
- [76] J.M. Gavilanes, T.C. Flatten, M. Brettel, Content strategies for digital consumer engagement in social networks: why advertising is an antecedent of engagement, *J. Advert.* 3367 (2018) 4–23, <https://doi.org/10.1080/00913367.2017.1405751>.

- [77] F. Sabate, J. Berbegal-Mirabent, A. Cañabate, P.R. Lebherz, Factors influencing popularity of branded content in Facebook fan pages, *Eur. Manag. J.* 32 (6) (2014) 1001–1011, <https://doi.org/10.1016/j.emj.2014.05.001>.
- [78] N. Kumar, L. Qiu, S. Kumar, A hashtag is worth a thousand words: an empirical investigation of social media strategies in trademarking hashtags, *Inf. Syst. Res.* (2022), <https://doi.org/10.1287/isre.2022.1107>.
- [79] Q. Liu, Y. Liang, S. Wang, Z. Huang, Q. Wang, M. Jia, Z. Li, W.K. Ming, Health communication through chinese media on e-cigarette: a topic modeling approach, *Int. J. Environ. Res. Public Health* 19 (13) (2022) 7591, <https://doi.org/10.3390/ijerph19137591>.
- [80] L. Xu, H. Lin, Y. Pan, H. Ren, J. Chen, Constructing the affective lexicon ontology, *Journal of the China Society for Scientific and Technical Information* 27 (2) (2008) 180–185.
- [81] Z. Dong, Q. Dong, “HowNet-a hybrid language and knowledge resource”, In *International conference on natural language processing and knowledge engineering, 2003. Proceedings. 2003*, Beijing, China 2003 (2003) 820–824, <https://doi.org/10.1109/NLPKE.2003.1276017>.
- [82] Y. Liu, Z. Duan, Chinese Movie Comment sentiment Analysis based on HowNet and User Likes, *J. Phys. Conf. Ser.* 1229 (1) (2019) 1, <https://doi.org/10.1088/1742-6596/1229/1/012018>.
- [83] P. Zhang, Z. Zhang, W. Zhang, C. Wu, Semantic similarity computation based on multi-feature combination using HowNet, *Journal of Software* 9 (9) (2014) 2461–2466, <https://doi.org/10.4304/jsw.9.9.2461-2466>.
- [84] R. Paternoster, R. Brame, P. Mazerolle, A. Piquero, Using the correct statistical test for the equality of regression coefficients, *Criminology* 36 (4) (1998) 859–866, <https://doi.org/10.1111/j.1745-9125.1998.tb01268.x>.
- [85] M. Bossetta, R. Schmøkel, Cross-platform emotions and audience engagement in social media political campaigning: Comparing candidates’ Facebook and Instagram images in the 2020 US election, *Polit. Commun.* 40 (1) (2023) 48–68, <https://doi.org/10.1080/10584609.2022.2128949>.
- [86] N.D. Albers-Miller, M.R. Stafford, An international analysis of emotional and rational appeals in services vs goods advertising, *J. Consum. Mark.* 16 (1) (1999) 42–57, <https://doi.org/10.1108/07363769910250769>.
- [87] E.E. Ko, D. Kim, G. Kim, Influence of emojis on user engagement in brand-related user generated content, *Comput. Hum. Behav.* 136 (2022) 107387, <https://doi.org/10.1016/j.chb.2022.107387>.
- [88] S. Zhang, M.D.T. de Jong, J.F. Gosselt, Microblogging for engagement: Effects of prior company involvement, communication strategy, and emojis on western and chinese users, *Journal of International Consumer Marketing* 34 (5) (2022) 616–630, <https://doi.org/10.1080/08961530.2022.2040073>.
- [89] S. Karray, L. Debernitz, The effectiveness of movie trailer advertising, *Int. J. Advert.* 36 (2) (2017) 368–392, <https://doi-org-ssl.oca.korea.ac.kr/10.1080/02650487.2015.1090521>.